RACE
Regular Analysis Through Continuous Exercise

Salt Analysis : RACE-01

Test of Carbonate and Bicarbonate

1. $CO_3^{2\ominus}$ and $HCO_3^{\ominus}$ can be distinguished by

(A) Phenolphthalein (B) $BaCl_2$ solution (C) $HgCl_2$ solution (D) All of these

CQA090

Test of Sulphite ($SO_3^{2\ominus}$); Bisulphite ($HSO_3^{\ominus}$)

2. A substance on treatment with dil. H_2SO_4 liberates a colourless gas which produces (i) turbidity with baryta water and (ii) turns acidified dichromate solution green. The reaction indicates the presence of :

(A) $CO_3^{2\ominus}$ (B) $S^{2\ominus}$ (C) $SO_3^{2\ominus}$ (D) $NO_2^{\ominus}$

CQA092

3. Which of the following options is correct

I. $HgCl_2$ can be used for the distinction of $HCO_3^{\ominus}$ and $CO_3^{2\ominus}$

II. Hg_2Cl_2 is insoluble in water

III. Aq. suspension of Ag_2CO_3 and Ag_2SO_3 both produces Ag_2O on heating

IV. SO_2 acts as reducing agent when passed through $FeCl_3$ solution

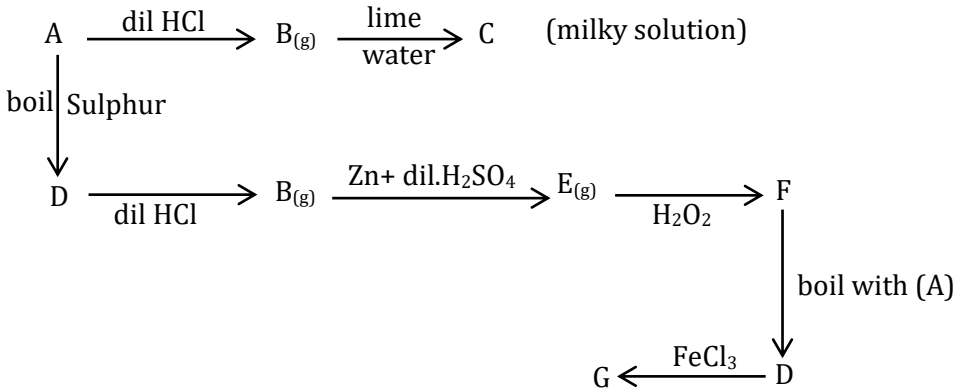
(A) FFTT (B) TFTT (C) TTFT (D) FTFF

CQA093

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Test of Thiosulphate ($S_2O_3^{2\ominus}$)

Paragraph for Q.4 to Q.5



4. Identify A

(A) $CO_3^{2\ominus}$ (B) $SO_3^{2\ominus}$ (C) $S_2O_3^{2\ominus}$ (D) None of these

CQA095

5. When A react with $Pb(NO_3)_2$ then compound X is formed. Compound X is oxidized by atmospheric oxygen on boiling, then Y is formed what is the colour of Y

(A) yellow (B) White (C) Black (D) Green

CQA096

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6. Match the column

Column-I	**Column-II**
(A) $S^{2\ominus}$	(P) Produces white turbidity of sulphur
(B) $HSO_3^{\ominus}$	(Q) Evolves gas with dil. HCl which turns lime water milky
(C) $SO_3^{2\ominus}$	(R) Evolves gas with dil.H_2SO_4 which does not turn Baryta water milky
(D) $S_2O_3^{2\ominus}$	(S) Colourless gas

(A) (A) → R; (B) → P; (C) → P,Q,S,R; (D) → P,S
(B) (A) → R,Q; (B) → Q; (C) → P,Q,S,R; (D) → P,S
(C) (A) → Q,S; (B) → Q,S; (C) → P,Q,S,; (D) → P,Q,S,R
(D) (A) → R,S; (B) → Q,S; (C) → Q,S; (D) → P,Q,S

CQA088

Test of Acetate ($CH_3COO^{\ominus}$)

7.

List–I (Molecule)	**List–II (Characteristic Odour)**
(P) CO_2	(1) Rotten egg smell
(Q) SO_2	(2) Suffocating smell of burning sulphur
(R) H_2S	(3) Vinegar like smell
(S) CH_3COOH	(4) Odour less

(A) P-4; Q-2; R-1; S-3 (B) P-2; Q-4; R-1; S-3
(C) P-3; Q-1; R-2; S-4 (D) P-2; Q-4; R-1; S-3

CQA089

8. Which of the following produces colourless gas with dil. H_2SO_4
(A) $Na_2S_2O_3$ (B) $NaHCO_3$ (C) $CH_3CO_2NH_4$ (D) All

CQA086

9. "Cacodyl oxide" is formed in the specific test of -
(A) Formate (B) Oxalate (C) Acetate (D) Nitrate

CQA002

Test of Chloride ($Cl^{\ominus}$), Bromide ($Br^{\ominus}$), Iodide ($I^{\ominus}$)

10. In layer test of $I^{\ominus}$ and $Br^{\ominus}$. If reddish -brown layer comes first then -
(A) $Br^{\ominus}$ present (B) $I^{\ominus}$ absent
(C) Both (A) and (B) (D) None of these

CQA100

11. Find the number of acidic radical(s) which can form coloured gas when treated with dil. H_2SO_4.
$CO_3^{2\ominus}$,$NO_2^{\ominus}$, $Br^{\ominus}$, $I^{\ominus}$,$SO_3^{2\ominus}$

CQA029

Test of Nitrate ($NO_3^{\ominus}$)

12. $NO_2^{\ominus}$ and $NO_3^{\ominus}$ can be distinguished by which of the following reagent.
(A) dil. H_2SO_4 (B) conc. H_2SO_4
(C) Devarda's alloy + conc. NaOH (D) None of these

CQA103

13. Brown ring complex bears the formula $[Fe(H_2O)_xNO]SO_4$ value of x will be :-

CQA028

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Paragraph for Q. 14 to Q. 15

$$B(g) \xleftarrow[H_2SO_4]{\text{Conc.}} \underset{\text{(Mixture of two ions)}}{(A)} \xrightarrow{\text{(Sulphamic acid)}} \underset{\text{(Only one Present)}}{(C)} \xrightarrow[+\text{conc.}H_2SO_4]{FeSO_4} (D)\ \text{(Unstable brown coloured compound)}$$

14. Identify gas B-

(A) Br_2 (B) $Br_2 + NO_2$ (C) NO_2 (D) None of these

CQA106

15. Nitrate is confirmed by ring test. The brown colour of the ring is due to formation of

(A) ferrous nitrite (B) nitroso ferrous sulphate

(C) ferrous nitrate (D) $FeSO_4NO_2$

CQA104

Test of Borate ($BO_3^{3\ominus}$)

16. Match the List-I and List-II

	List–I (Acidic radicals)		**List–II (Test)**
(P)	$NO_2^{\ominus}$	(1)	Green flame test
(Q)	$BO_3^{3\ominus}$	(2)	Cacodyl oxide reaction
(R)	$Br^{\ominus}$	(3)	Griess Ilosvay test
(S)	$CH_3COO^{\ominus}$	(4)	Layer test

Code :

	P	Q	R	S
(A)	4	2	1	3
(B)	3	1	4	2
(C)	3	4	2	1
(D)	4	3	2	1

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CQA005

Test of Oxalate ($C_2O_4^{2\ominus}$)

17. Salt of which of the following radicals will give brown fumes when treated with conc. H_2SO_4 :-

(A) $NO_3^{\ominus}$ (B) $C_2O_4^{2\ominus}$ (C) $I^{\ominus}$ (D) $Cl^{\ominus}$

CQA102

18. Which compound will not give positive chromyl chloride test?

(A) Copper chloride, $CuCl_2$ (B) Mercuric chloride, $HgCl_2$

(C) Zinc chloride, $ZnCl_2$ (D) Sodium chloride, NaCl

CQA101

19. Which of the following acidic radical form coloured ppt. (other than white) with $AgNO_3$ solution.

$Cl^{\ominus}, I^{\ominus}, CrO_4^{2\ominus}, NO_2^{\ominus}, C_2O_4^{2\ominus}, S^{2\ominus}$

CQA024

Test of Sulphate ($SO_4^{2\ominus}$)

20. A White solid is first heated with dil. H_2SO_4 and then with conc. H_2SO_4. No action was observed in either case. The solid salt contains

(A) Sulphide (B) Sulphite (C) Thiosulphate (D) Sulphate

CQA015

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	C	B	B	D	A	D	C	C
Question	11	12	13	14	15	16	17	18	19	20
Answer	1	A	5	C	B	B	A	B	3	D

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RACE
Regular Analysis Through Continuous Exercise

Salt Analysis : RACE-02

Classification of Basic Radicals

1.

List–I (Cations)		List–II (Group reagent)	
P	$Co^{2\oplus}$	1	$(NH_4)_2CO_3$ in presence of NH_4Cl & NH_4OH
Q	$Fe^{3\oplus}$	2	H_2S gas in acidic medium (dil HCl)
R	$Cu^{2\oplus}$	3	H_2S in presence of NH_4OH & NH_4Cl
S	$Ca^{2\oplus}$	4	NH_4OH in presence of NH_4Cl

(A) P-4; Q-2; R-1; S-3 (B) P-3; Q-1; R-4; S-2
(C) P-1; Q-2; R-3; S-4 (D) P-3; Q-4; R-2; S-1

CQA116

Test of Basic Radicals : Group 1

2. $Hg_2I_2\downarrow$ (green) $\xrightarrow[\text{with } H_2O]{\text{boiled}}$ products

Which of the following statement is / are correct with respect to the products ?
(A) Black precipitate of mercury(I) oxide is formed.
(B) Violet colour gas is evolved.
(C) Red precipitate of HgI_2 is formed.
(D) Mercury is obtained

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CQA058

Test of Basic Radicals: Group 2B

3. Three separate samples of a solution of a single salt gave these results. One formed a white precipitate with excess ammonia solution, one formed a white precipitate with dilute NaCl solution and one formed a black precipitate with H_2S. The salt could be :
(A) $AgNO_3$ (B) $Pb(NO_3)_2$ (C) $Hg(NO_3)_2$ (D) $Mn(NO_3)_2$

CQA117

4. An aqueous solution of a substance gives a white ppt. on treatment with dil. HCl, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black ppt. is obtained. The substance is a
(A) $Hg^{2\oplus}$ salt (B) $Cu^{2\oplus}$ salt (C) $Ag^{\oplus}$salt (D) $Pb^{2\oplus}$salt

CQA004

5. The total number of species which give black precipitate with H_2S
$Hg(NO_3)_2$, $CuSO_4$, $AgNO_3$, $Pb(NO_3)_2$, $Sn(NO_3)_2$

CQA023

Test of Basic Radicals : Group 3

6. In the precipitation of the iron group in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to
(A) decrease concentration of $OH^{\ominus}$ ions. (B) prevent interference by phosphate ions.
(C) increase concentration of $Cl^{\ominus}$ ions. (D) increase concentration of $NH_4^{\oplus}$ ions.

CQA124

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7. $FeSO_4 + KCN$ (excess) $\xrightarrow{\Delta}$ (X) → Conc. H_2SO_4 → (Y) gas ↑; dil. H_2SO_4 → (Z) gas ↑

The sum of number of pπ-pπ bonds present in gas (Y) & (Z) ?

CQA068

8. Few drops of HNO_3 are added to II group before proceeding to III group in order to :-

(A) Convert $Fe^{2\oplus}$ to $Fe^{3\oplus}$ (B) Convert $Fe^{3\oplus}$ to $Fe^{2\oplus}$

(C) ppt. III group (D) None of these

CQA016

Test of Basic Radicals : Group 4

9. To avoid the precipitation of Hydroxides of $Zn^{2\oplus}$, $Mn^{2\oplus}$ and $Ni^{2\oplus}$ along with those of $Fe^{3\oplus}$, $Al^{3\oplus}$ and $Cr^{3\oplus}$ the third group solution should be :

(A) Concentrated HNO_3 is added (B) Treated with excess of NH_4Cl

(C) Concentrated H_2SO_4 is added (D) Treated with excess of NH_4OH

CQA125

10. An aqueous solution of colourless metal sulphate M gives a white precipitate with NH_4OH. This was soluble in excess of NH_4OH. On passing H_2S gas through this solution a white precipitate is formed. The metal M in the salt is :

(A) Ca (B) Ba (C) Al (D) Zn

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CQA129

11. The auxiliary reagent in group-IV reagent is

(A) H_2S (B) dil. HCl (C) NaOH (D) NH_4OH

CQA006

12. Find the number of radical which can not form white coloured precipitate with $K_4[Fe(CN)_6]$.

$Fe^{3\oplus}$, $Fe^{2\oplus}$, $Cu^{2\oplus}$, $Zn^{2\oplus}$, $Ca^{2\oplus}$

CQA021

Test of Basic Radicals : Group 5

13. A metal salt solution form a yellow ppt with potassium chromate in acetic acid, a white ppt with dilute sulphuric acid but gives no ppt with sodium chloride or iodide it is :

(A) Lead carbonate (B) Basic lead carbonate

(C) Barium nitrate (D) Strontium nitrate

CQA131

14. The solid laboratory reagent 'A' gives the following reactions -

(i) It imparts green colour to the flame

(ii) Its solution does not give a precipitate on passing H_2S

(iii) When it is heated with solid $K_2Cr_2O_7$ and concentration H_2SO_4, a red gas is evolved. When this gas passed into aqueous solution of NaOH, turns it yellow -

Identify 'A'

(A) $PbCl_2$ (B) $BaCl_2$ (C) NaCl (D) None of these

CQA132

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15. How many chromates are yellow precipitate ?
$BaCrO_4$; $SrCrO_4$; $PbCrO_4$; Hg_2CrO_4 ; Ag_2CrO_4

CQA026

16. In how many of the following reactions, one of the product is obtained as a yellow precipitate :

(a) $Ba^{2\ominus} + CrO_4^{2\ominus} \longrightarrow$ product

(b) $NH_4^{\oplus} + [PtCl_6]^{2\ominus} \longrightarrow$ product

(c) $NH_4^{\oplus} + [Co(NO_2)_6]^{3\ominus} \longrightarrow$ product

(d) $Ca^{2\oplus} + [Fe(CN)_6]^{4\ominus} + K^{\oplus} \longrightarrow$ product

(e) $Sr^{2\oplus} + CO_3^{2\ominus} \longrightarrow$ product

CQA069

Test of Basic Radicals : Group 6

17. A metal 'X' on heating in nitrogen gas gives 'Y'. 'Y' on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour Y is:

(A) $Mg(NO_3)_2$ (B) Mg_3N_2 (C) NH_3 (D) MgO

CQA130

18. Number of species which show white precipitate out of the following :-
SnS_2, SnS, CdS, CuS, PbS, $Al(OH)_3$, AgCl, ZnS, ZnO

CQA030

Charcoal Cavity test, Cobalt Nitrate Charcoal Cavity Test

19. The compound present in borax bead is

(A) B_2O_3 (B) $NaBO_2$ (C) $NaBO_3$ (D) $NaBO_2 + B_2O_3$

CQA113

Heating Effect

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20. **Match the column**

Column-I	**Column-II**
(A) $Be(NO_3)_2 \xrightarrow{\Delta}$	(P) Gives H_2O
(B) $HNO_2 \xrightarrow{\Delta}$	(Q) Oxyacid is obtained
(C) $NaH_2PO_4 \xrightarrow{\Delta}$	(R) Gives disproportionation reaction
(D) $H_3PO_3 \xrightarrow{\Delta}$	(S) Oxygen gas is evolved

(A) (A→P, B→Q, C→R, D→S) (B) (A→S, B→P,Q,R, C→P, D→Q,R)
(C) (A→P, B→Q, C→S, D→R) (D) (A→Q, B→P, C→S, D→R)

CQA109

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	CD	B	D	4	A	4	A	B	D
Question	11	12	13	14	15	16	17	18	19	20
Answer	D	2	C	B	3	3	B	4	D	B

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